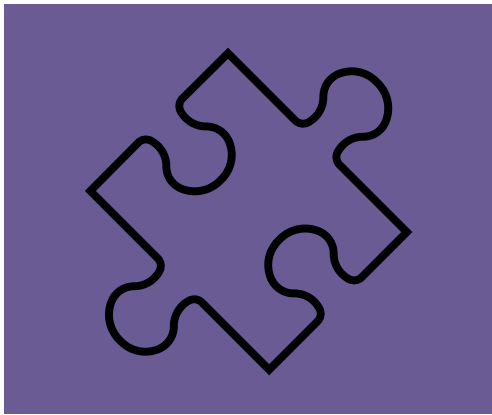
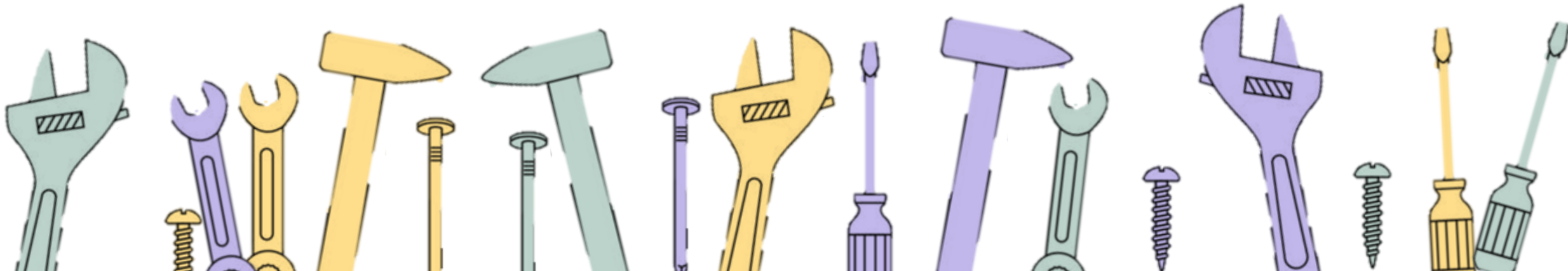


Horses for courses – The Game



Statistics is not about memorising tests.
It's about recognising structure.



Rules of engagement

-  **It's a game — and games are voluntary.** If you prefer to work alone or prefer not to actively play, that's completely fine. You're welcome to observe, think quietly, or contribute in your own way.
-  **Justify before you stick.** You must explain your reasoning to your group before placing a card.
-  **Read the board.** Some of your tools and visualisations belong to other groups' cases.
-  **Challenge ideas, not people.** If you disagree, explain *why* using the decision tree.
-  **You may challenge one placement from another group.** But you must provide a clear statistical argument.

Horses for courses – The Game

There are

- 12 Case cards
- 12 Tool cards
- 12 Visualisation cards
- 4 Groups

Each group receives:

- 3 Case cards
- 3 Tool cards
- 3 Visualisation cards

Your group's mission is to:

- 1) Place your **case cards** on the correct board (Inference or Probability)
- 2) Identify the correct **data structure**
- 3) Match the correct **statistical tool**
- 4) Assign the most appropriate **visualisation**

But there's a twist...

You only hold part of the puzzle.

Some of your tools and visualisations belong to other groups' cases — so you'll need to read the boards carefully, challenge decisions, and think like statisticians.

The goal:

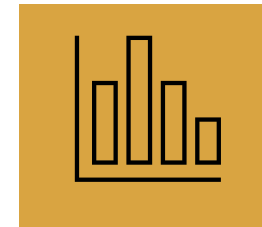
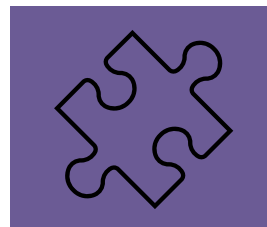
👉 Make sure every case ends up with the correct method and visualisation — and be ready to defend your choices.

Describe the data

Choose the right tool.

Pick the right visualisation.

Defend your reasoning.



Are you calculating a probability or analysing collected data?

PROBABILITY PATH

1. Is there a fixed number of repeated trials with two outcomes (success/failure)?

- **Binomial distribution**

- Exact number → Exact binomial probability
- “Or more extreme” → Tail probability

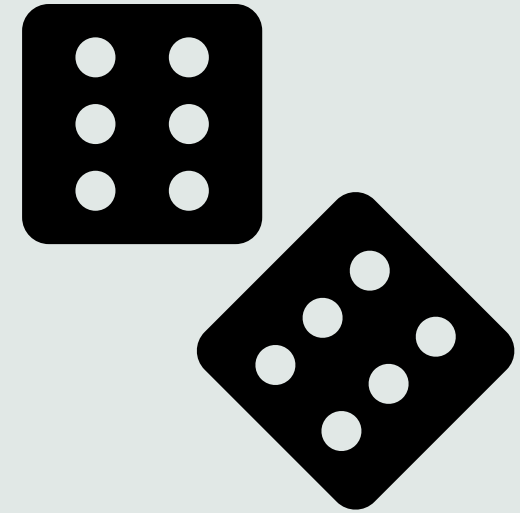
2. Are you combining events?

“A OR B”

- Are events mutually exclusive?
 - Yes → **Addition rule**
 - No → Addition minus overlap

“A AND B”

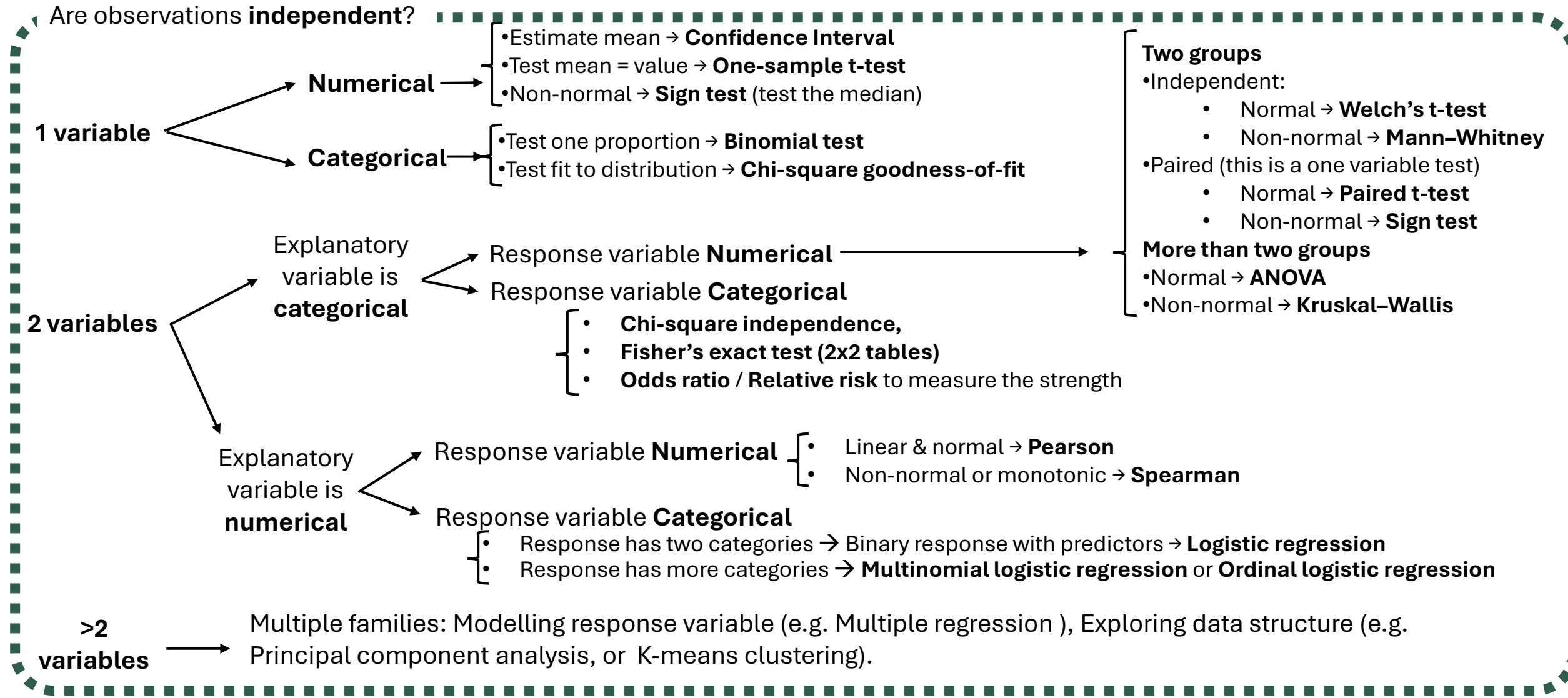
- Are events independent?
 - Yes → **Multiplication rule**
 - No → **Conditional probability**



For analysing collected
data check next diagram

HOW TO CHOOSE A MODEL

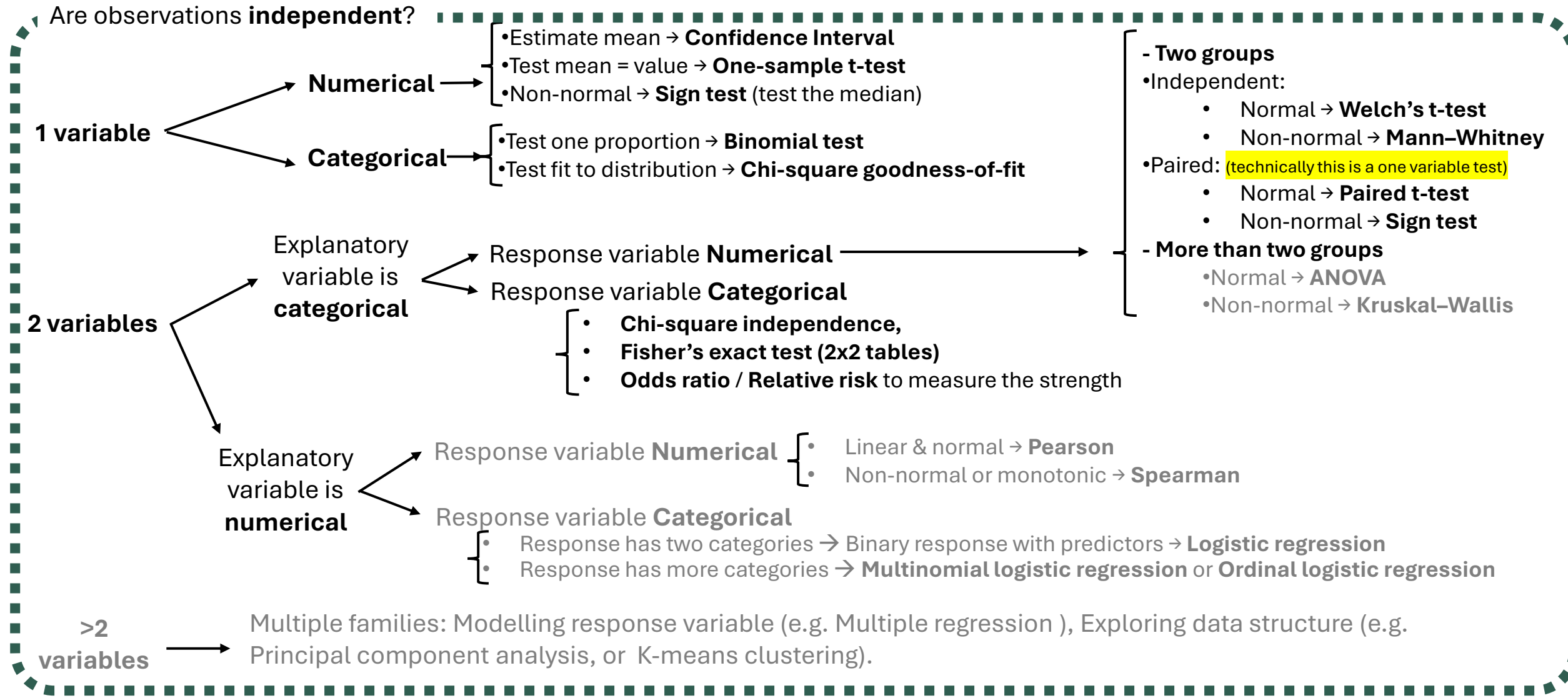
(test-by-test way of thinking about statistics)



Are observations **not independent**? → No (repeated measures, clustering, nested data) → Use a **Mixed Model (LMM, GLMM)**

HOW TO CHOOSE A MODEL

(test-by-test way of thinking about statistics)



HOW TO CHOOSE A MODEL

(modelling version)

If response is:

- Numerical → Linear Model (GLM)
- Binary → Binomial GLM (Logistic regression)
- Count → Poisson GLM
- Hierarchical → Mixed Model (LMM / GLMM)

Meta-learning exercises

Which distinction caused the most confusion today?

- Probability vs inference?
- GOF vs independence?
- Paired vs independent?
- Normal vs non-normal?

Before choosing a method, always identify:

- 1.One variable or two?
- 2.Numerical or categorical response?
- 3.Independent or paired?
- 4.Normal or not?
- 5.Probability or inference?

If you can answer those, the method follows.

For you to practice: Check that you understand the difference between:

- Binomial test vs χ^2 GOF (Case 2 vs Case 3)
- Probability vs p-value thinking (Case 9 vs inference cases)
- Welch vs ANOVA
- Mann–Whitney vs Welch
- Independence vs multiplication rule (Case 7 vs Case 12)

Suggested consolidating exercise: write down your own example cases on the decision tree.

SOLUTIONS

Case	Research question	Data structure (One variable / Two variables; Numerical or Categorical; Independent or Paired; # Groups)	Key assumption	Statistical method	Visualisation
1	Is mean mass different from 10 g?	One variable; Numerical; One sample	Approx. normal	One-sample t-test	Histogram + mean & CI
2	Is germination rate different from 50%?	One variable; Categorical (success/failure); One sample	Independent trials; binomial setting	Binomial test	Bar plot (proportion)
3	Do parasite counts follow a Poisson distribution?	One variable; Categorical (counts in categories)	Expected frequencies adequate	Chi-square goodness-of-fit	Observed vs expected bar plot
4	Do treatments differ in mean height?	Two variables; Numerical response; Categorical explanatory (2 groups); Independent	Approx. normal; unequal variance	Welch's t-test	Boxplot (2 groups)
5	Do treatments differ in recovery time?	Two variables; Numerical response; Categorical explanatory (2 groups); Independent	Non-normal distribution	Mann–Whitney U	Boxplot / violin
6	Does treatment change blood pressure?	Two variables; Numerical response; Paired measurements	Approx. normal differences	Paired t-test (Sign test if non-normal)	Paired dot plot
7	Is disease associated with smoking?	Two variables; Both categorical; Independent; 2×2 table	Expected counts adequate	Chi-square independence (+ OR/RR)	Mosaic plot
8	Do means differ among temperatures?	Two variables; Numerical response; Categorical explanatory (>2 groups); Independent	Approx. normal	ANOVA (Kruskal–Wallis if non-normal)	Bar plot with error bars

Case	What are we calculating?	Describe the situation (categories, events, mutually exclusive or not)	Rule / Distribution to use	Visual representation
9	Probability that exactly 2 out of 5 plants are infected	Repeated independent trials; fixed number of trials (5); two outcomes per trial (infected / not infected); same probability each time	Binomial distribution (exact probability)	Probability mass function (bar plot of binomial probabilities)
10	Probability the bird eats insects OR fruit	Single event with multiple categories; categories are mutually exclusive (only one food type per feeding event)	Addition rule (for mutually exclusive events)	Bar chart or simple categorical diagram
11	Probability an infected individual was vaccinated (P(Vaccinated Infected))	Two related events; not mutually exclusive; conditional relationship; requires reversing conditioning	Conditional probability (Probability tree / Bayes reasoning)	Probability tree diagram
12	Probability offspring has both dominant traits (A AND B)	Two separate events; events are independent; joint occurrence required	Multiplication rule (independent events)	Tree diagram showing branches multiplied

(for teachers) Elements to deliver to each group:

GROUP 1

Cases:

- 1 (One-sample t-test)
- 7 (Chi-square independence)
- 9 (Binomial probability)

Tools:

- 1 (One-sample t-test)  matches Case 1
- 5 (Mann–Whitney)
- 11 (Probability tree)

Visualisations:

- 1 (Histogram + CI)  matches Case 1
- 7 (Mosaic plot)
- 10 (Bar/pie chart)

They can correctly solve Case 1 themselves.

They must look at the board to place:

- Mosaic plot
- Probability tree
- Mann–Whitney

GROUP 2

Cases:

- 2 (Binomial test)
- 4 (Welch t-test)
- 10 (Addition rule probability)

Tools:

- 2 (Binomial test)  matches Case 2
- 8 (ANOVA)
- 12 (Multiplication rule)

Visualisations:

- 2 (Bar plot proportion)  matches Case 2
- 4 (Boxplot 2 groups)
- 12 (Tree diagram independent)

They can correctly solve Case 2 themselves.

Must read board for:

- Welch t-test
- Addition rule

(for teachers) Elements to deliver to each group:

GROUP 3

Cases:

3 (Chi-square GOF)

6 (Paired t-test)

11 (Conditional probability)

Tools:

6 (Paired t-test)  matches Case 6

3 (Chi-square GOF)

9 (Binomial formula)

Visualisations:

6 (Paired dot plot)  matches Case 6

3 (Observed vs expected bars)

11 (Probability tree)

They can correctly solve Case 6.

But they have correct tool for Case 3 — so that's interesting:

They'll realise they can solve two of their own cases.

Still must read board for conditional probability.

GROUP 4

Cases:

5 (Mann–Whitney)

8 (ANOVA)

12 (Multiplication rule probability)

Tools:

4 (Welch t-test)

7 (Chi-square independence)

10 (Addition rule)

Visualisations:

5 (Boxplot skewed)

8 (Boxplot >2 groups)

9 (Binomial PMF)

This group initially has ZERO direct matches.

They must rely entirely on reasoning + checking other groups.

This creates asymmetry — which is good.

Analysing Data Paths:

CASE

RESEARCH QUESTION

DATA STRUCTURE

KEY ASSUMPTION

STATISTICAL METHOD

VISUALISATION

Is the claim that the new drug is superior to the old one supported by this sample?

One variable: numerical

Assumes normal distribution

Two-sample t-test

Bar chart

Is there an association between smoking and prostate size?

2 variables

Chi-squared test

Contingency table

Bar chart

How does temperature affect enzyme activity?

2 variables

Assumes normal distribution

Line graph

Bar chart

Do people with a higher IQ score have a higher income?

One variable: numerical

Assumes normal distribution

Scatter plot

Bar chart

Which treatment (A or B) produces faster speed of recovery?

2 variables

Assumes independence and normality

Two-sample t-test

Bar chart

Do plants grow more under UV or visible light?

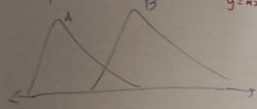
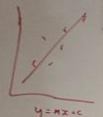
2 variables

Assumes independence and normality

Two-sample t-test

Bar chart

#	Treat	Re T
1	A	✓
2	A	✓
3	B	✓
4	B	✓



Probability:

CASE

DATA DESCRIPTION

WHAT ARE WE CALCULATING?

RULE / DISTRIBUTION

VISUALISATION

Mutually exclusive

Probability of bird eating either of two types of seed

Tree diagram

Bar chart

Not mutually exclusive

Probability of a person having both a car and a dog

Tree diagram

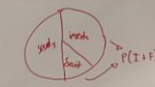
Bar chart

Categories

Proportion of infected individuals that are vaccinated

Tree diagram

Bar chart



$$P(AB) = 0.5 \times 0.4 = 0.2$$

